

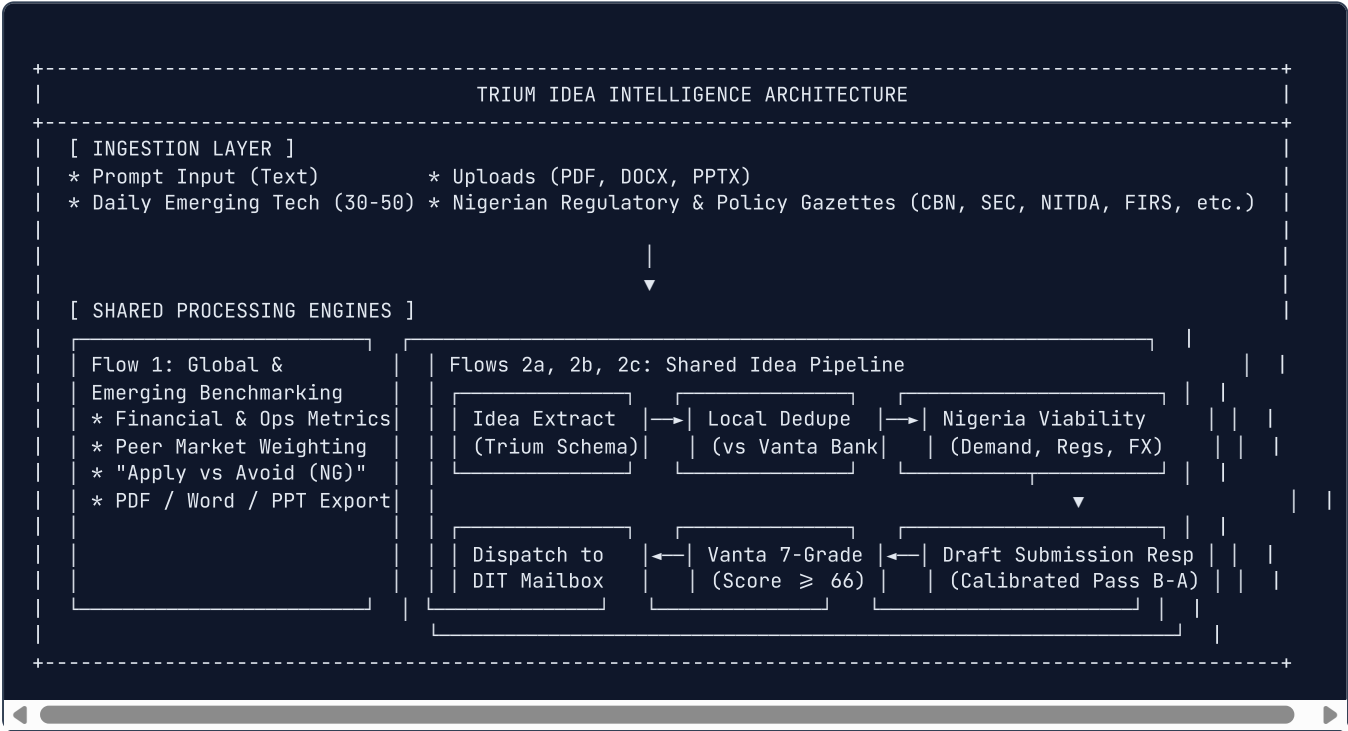
Trium Idea Intelligence Platform (Project Reva)

Product Requirements Document — Global Benchmarking, Local Viability & Daily Idea/Policy Scouting

DOCUMENT VERSION	PRODUCT OWNER	PRIMARY RECIPIENT	SOFTWARE COST
v1.1.0 (Incorporated)	Olanrewaju Taiwo	"Digital Incubation" Team	\$0.00 / Free Tools Only

1. Executive Summary & Strategic Context

Trium Limited requires a unified platform that performs two vital venture building jobs: First, it benchmarks candidate ideas against what has been built globally, with deliberate emphasis on emerging and nearby African markets, generating actionable **Apply in Nigeria** and **Avoid in Nigeria** operational guidance. Second, it screens ideas for Nigerian market viability, compares them with the **Vanta Idea Bank** using privacy-preserving local deduplication, pre-grades candidate initiatives against Trium's established 7 criteria, and delivers them either on-demand or automatically every morning to the **Digital Incubation Team (DIT)**.



2. The Four In-Scope Flows

FLOW	WHAT IT DOES	TRIGGER	OUTPUT & DESTINATION
1. Global Benchmarking	Researches an idea across global, emerging and nearby markets; extracts comparable operational numbers, business models, and failure post-mortems.	On demand (prompt or doc upload)	Report in PDF, Word (DOCX) or PowerPoint (PPTX) with dedicated Apply / Avoid guidance for Nigeria.
2a. Local Viability	Extracts ideas from prompt/document, removes exact duplicates in Vanta, tests survivors for Nigerian viability, drafts and grades submission responses.	On demand	Graded, submission-ready entries with full Vanta scorecard and gap breakdown. Exportable as PDF/Word/PPT.
2b. Emerging Tech Scout	Scrapes 30 to 50 emerging-market tech sources daily (Nigeria data excluded; global fallback to Tier B if <30); runs shared 2a pipeline.	Daily (05:00 WAT cron)	Automated structured HTML email to digital-incubation@trium.ng for ideas achieving Grade B or higher.
2c. Nigerian Policy Scout	Scrapes Nigerian regulatory bodies, official gazettes, and legal trackers for incoming and enacted policy; derives venture opportunities; runs 2a pipeline.	Daily (06:00 WAT cron)	Automated email to digital-incubation@trium.ng for ideas achieving Grade B or higher, citing the governing policy clause.

3. Vanta Integration & Established 7–Criteria Scale

Vanta is the single source of truth for the Idea Bank and evaluation rules. In Version 1.0, **Vanta access is read-only** (read via API GET /api/v1/portfolio or direct Convex role). Automatic write-back into Vanta is deferred to a future phase after human sign-off.

CRITERION	WEIGHT	KEY	EVALUATION CONSIDERATION & CALIBRATION
1. Strategic Alignment	20 pts	strategic_alignment	Adheres to Trium ideation themes; aligns with long-term studio vision; leverages Trium's strategy wheel and unfair capabilities.
2. Customer–Problem Fit	20 pts	customer_problem	Solves a real, acute, and verified customer/partner pain point in Nigeria; validated willingness to adopt and repeatedly pay.
3. Solution Fit	15 pts	solution_fit	Expands Trium customer base; sizable addressable market (TAM/SAM); deep understanding of market dynamics to secure market leadership.
4. Market Opportunity	15 pts	market_opportunity	Unique versus incumbents and informal alternatives; significantly improves broken supply chains; creates high economic impact.
5. Differentiation & Moat	10 pts	differentiation	Long-term relevance; defensibility against well-funded incumbents, copy-cats, platform disruption, and disintermediation.
6. Sustainable Advantage	10 pts	sustainable_advantage	Trium studio capability fit; resource and capital availability; access to proprietary distribution; mitigation of macro/FX headwinds.
7. Feasibility & Scalability	10 pts	feasibility	Expandability into ancillary products/fintech rails; ease of geographic expansion across Nigerian geopolitical zones and regional Africa.

Vanta Scoring Scale & Action Mapping:

A* (86–100 pts) Outstanding — Immediate IC Fast-Track |
 A (76–85 pts) Excellent — Pass & Advance to EVO Gate 0/1 |
 B (66–75 pts) Good — **PASS THRESHOLD & DIT TRIGGER** |
 C (57–65 pts) Average — Hold in Bank / Needs Validation |
 D (<57 pts) Below Threshold — Decline

4. Confidentiality & Local Model Architecture (Zero Data Exposure)

Client Requirement on Data Privacy: *"sensitive steps should use a locally run open model or an efficient alternative that don't expose data."*

To guarantee that unpublished, confidential ideas inside the Vanta Idea Bank are never leaked to external public AI providers whose terms allow training:

- **Local Deduplication Embeddings:** High-dimensional embeddings of Vanta's Idea Bank are computed **100% on-device** using local open-source models (sentence-transformers/all-MiniLM-L6-v2). No proprietary Vanta

text is sent to third-party embedding APIs.

- **Candidate Shortlisting:** Nearest-neighbor cosine similarity runs locally in Python/pgvector to identify candidate pairs.
- **Local LLM Judge:** For ambiguous candidate pairs (similarity between 65% and 88%), Reva uses a **local open model** (via local Ollama / Llama 3.2 3B) or an on-device semantic differencing rule to determine whether an idea is an *Exact Duplicate*, *Near-Similar* (differentiated wedge), or *New*.
- **Public Data Pipeline:** Public benchmark scanning (Flow 1) and public news/policy scraping (Flows 2b and 2c) consume zero internal company data and safely utilize the generous Google Gemini 2.0 Flash Free Tier and Groq Free Tier at **\$0.00 cost**.

5. Governance & Source Registry Rules

Dual Sign-Off Governance Policy:

As established by Trium, any update to the curated source registries (emerging tech portals or Nigerian regulatory trackers) or monthly additions proposed by the autonomous discovery engine **requires explicit sign-off from both a Reva Platform Admin and a Vanta Admin** before entering active daily production.

Flow 2b: Emerging Market Sources (35 Tier A + 15 Tier B)

- **Africa (Ex-NG):** Disrupt Africa, WeeTracker, TechMoran, Techweez, Ventureburn, CIO Africa, Tech In Africa, KokoFeed.
- **MENA:** Wamda, MAGNiTT, Enterprise Egypt, Waya Media.
- **Southeast Asia:** Tech in Asia, e27, DealStreetAsia, KrASIA, DailySocial ID, Vulcan Post, Vietcetera.
- **South Asia:** Entracker, Inc42, YourStory, MediaNama.
- **Latin America:** Contxto, Startups Real, Startups.com.br, Tekios, LatamList.
- **Tier B Global Fallbacks:** Crunchbase News, TechCrunch, VentureBeat, Sifted, PitchBook, Rest of World.

Flow 2c: Nigerian Regulatory & Policy Watch

- **Scope:** Enacted circulars AND incoming instruments (National Assembly proposed bills, regulator exposure drafts, consultative papers).
- **Regulators:** Central Bank of Nigeria (CBN), SEC Nigeria, NCC, NITDA, FCCPC, FIRS, NAFDAC, NERC, NAICOM, FMCIDE.
- **Official Records:** Federal Republic of Nigeria Official Gazettes.
- **Policy Journalism:** BusinessDay Law & Policy, Nairametrics Regulatory Watch, Stears, TechCabal Policy Watch.

6. Digital Incubation Team (DIT) Notification Specification

- **Official Recipient:** "Digital Incubation" <digital-incubation@trium.ng>
- **Email Engine:** Delivery via **Resend** (leveraging Trium/Vanta's existing Resend account) with SMTP failover.
- **Trigger Rule:** Automated immediate dispatch whenever an idea achieves **Grade B, A, or A* (Score ≥ 66)** during daily 2b or 2c runs.
- **Email Brief Content:**
 - *Subject Line:* [TRIUM INTEL] [Grade {Grade} - {Score}/100] {Idea Name} – {Origin Market / Regulatory Policy}
 - *Executive Overview:* Distinctive idea name, core problem, proposed solution, target customer segment, monetization model.
 - *Origin Benchmark or Policy Clause:* Specific startup/model observed abroad OR exact regulatory section/bill cited.
 - *The Nigerian Play:* Specific Apply and Avoid recommendations tailored to local payment rails, infrastructure, and consumer willingness-to-pay.
 - *Vanta Scorecard Breakdown:* Scores across all 7 criteria with concise rationales, 3 Key Strengths, and 3 Key Risks.
 - *Review Action:* Direct links to review full benchmark or park idea.

7. Free-Tier Tool Selection Matrix (Strict \$0 Mandate)

TOOL CATEGORY	TOOL SELECTED	FREE TIER QUOTA & FIT	ARCHITECTURAL FUNCTION
Language Model & Reasoning	Google Gemini 2.0 Flash	100% Free (15 RPM, 1,500 RPD, 1M context)	Primary LLM for public document extraction, benchmark research, and drafting.
	Groq Cloud (Llama 3.3 70B)	100% Free (30 RPM, 14.4k RPD)	Fallback fast LLM for article classification and JSON sanitization.
Privacy Deduplication	Sentence-Transformers (Local)	100% Free & Unlimited (On-device compute)	Computes Vanta Idea Bank embeddings locally with zero data sent outside.
Scraping & Feed Extraction	Feedparser & Trafilatura	100% Free Open Source (Local Python)	Core daily scraper for RSS feeds, sitemaps, and clean article body text.
	Playwright / Crawl4AI	100% Free Open Source (Local Headless)	Scrapes dynamic JavaScript pages without burning third-party API credits.
Targeted Search	Tavily Search API	Free Tier: 1,000 searches/month	Targeted research queries for Flow 1 benchmarks and monthly discovery.
	Exa Search API	Free Tier credits	Semantic company search and finding parallel emerging market peers.
Document Parsers	pdfplumber, python-docx, pptx	100% Free Open Source	Local extraction of tables, text, and layouts from uploaded documents.

8. Phased Delivery Plan

PHASE	SCOPE & DELIVERABLES	TOOLING USED	STATUS
Phase 0 & 1	Discovery & Flow 1: Global Benchmarking * Confirm Vanta read schemas, grading scale & criteria. * Input ingestion (prompt, PDF, DOCX, PPTX). * Multi-tier benchmark research (Global, Emerging, Nearby). * "Apply vs Avoid (Nigeria)" generation & Multi-format export (PDF, Word, PPT).	Python 3.12, pdfplumber, python-pptx, Gemini 2.0 Flash Free Tier, Tavily Free Tier.	Workspace Ready PRD Approved
Phase 2	Shared Pipeline & Flow 2a: Local Viability * Vanta read connector (Idea Bank sync). * Local sentence-transformers deduplication (zero data exposure). * Nigeria viability assessment (Medium+ gate). * Trium 7-criteria response drafting calibrated for Grade B-A.	Vanta HTTP API, sentence-transformers (local), Gemini Flash, Pydantic schemas.	Scheduled Next
Phase 3	Autonomous Daily Scouts (Flow 2b & 2c) * Curated 50-source emerging tech feed crawler (ex-NG, Tier B fallback). * Nigerian regulatory, exposure draft & gazette crawler. * Daily cron runners (05:00 and 06:00 WAT). * Automated Resend email dispatcher to digital-incubation@trium.ng.	feedparser, trafilatura, Playwright, Resend API, Jinja2 templates.	Scheduled Next
Phase 4	Hardening & Governance * Dual admin sign-off dashboard for source additions. * Monthly AI discovery runner for emerging tech trackers. * End-to-end integration testing with live Vanta assessments.	FastAPI, SQLite/pgvector, AntigraVity Agent.	Future Iteration